

Aditya Balaji

Student researcher and software engineer | Computer Science + quantitative finance

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Cambridge student building research systems and data-driven software at the intersection of markets, machine learning, and reliable engineering. Corresponding author on a peer-reviewed Q1 finance paper; develops tools used by others for analysis, simulation, and decisions.

EDUCATION

JBCN International School, Mumbai - Cambridge International (CAIE)

Expected 2027

- A Levels: Mathematics, Further Mathematics, Computer Science, Economics, Physics, English General Paper; IGCSE: 93%.
- Cambridge Independent Research Report (IPQ 9980), 4,982 words: investigated whether high-frequency trading destabilises markets during stress using an independently designed research question and empirical analysis.

RESEARCH

Global Attention Saturation Index (GASI)

Lead / corresponding author - Borsa Istanbul Review (Q1), peer-reviewed

- Measured investor-attention thresholds across 22 international equity markets using Lewbel IV identification to address endogeneity between attention and volatility.
- Estimated a structural threshold of $\gamma^* = 1.5625$ by profile likelihood (Hansen set [1.14, 2.50]); documented the Silence Signature, confirmed in 14 of 22 markets through synthetic-control falsification with Driscoll-Kraay standard errors.

Behavioral Ownership Gap in Fantasy Premier League

Corresponding author - Journal of Sports Economics (Q1), under review

- Formalised the gap between optimal and actual ownership in a GBP 7.5B ecosystem; a Fama-MacBeth panel across 380 gameweeks returned $t = -57.20$.
- Estimated a GBP 137.1M annual welfare cost from information-architecture failure; a frozen, out-of-sample long-short replication produced 105 profitable weeks out of 105.

PhenoSync - phenological mismatch and climate value-at-risk

Sole author - Oxford Said Climate Challenge

- Joined GBIF, NASA MODIS (LST/NDVI), NOAA GHCND, and FAOSTAT data to quantify a 4.57-day/year phenology shift ($R^2 = 0.83$, $p = 0.0003$) across a seven-framework robustness battery.
- Designed a PMI-severity crop-exposure model translating ecological mismatch into interpretable, billion-dollar value-at-risk indicators.

Why Overparameterised Neural Networks Generalise

Independent mathematical exposition

- Derived the Neural Tangent Kernel from first principles using Further Mathematics, connecting gradient dynamics, spectral bias, and generalisation through a fully worked two-neuron example.

ENGINEERING

ATLAS (QUANT360) - quantitative research platform

Solo build - in production use

- Built eight research modules spanning regimes, screening, ML, volatility, factors, PCA, and portfolios; made the platform usable for independent analysis by another CS graduate.
- Implemented causal engines for DiD, RD, IV, and event studies; a volatility suite for HAR-RV, GARCH, jumps, and connectedness; and a bias-aware ML stack with purged CV, deflated Sharpe, Double ML, meta-labelling, and hierarchical risk parity.

NEXUS Exchange - agent-based market simulator

Developer / researcher - SSRN paper

- Simulated nine behaviourally distinct trader archetypes across 10,200 Monte Carlo runs with Numba and multiprocessing; recovered $\gamma^* = 1.5625$ by Simulated Method of Moments, independently corroborating GASI.
- Validated nine canonical stylised facts with jump-filtered ARCH-LM tests and shipped a five-tier FastAPI architecture with a PPO adversarial market maker and XGBoost/LSTM fragility predictor.

Glassbox SRE - autonomous incident response

Developer

- Connected Prometheus/Alertmanager, LangGraph investigation, parallel commit correlation, pgvector runbook retrieval, and queried impact estimation into an evidence-cited incident brief.
- Built a 15-scenario evaluation harness with seeded ground-truth commits: 86.7% top-1 accuracy versus a 13.3% deterministic baseline.

ENGINEERING - CONTINUED

Daedalus - engineering decision platform

Solo developer - used in STEM Racing

- Built a unit- and dimension-validated execution engine with provenance tracking, Monte Carlo sensitivity, multi-objective Pareto optimisation, and equation traceability across Next.js, FastAPI, PostgreSQL, and Three.js.

Zeus - agentic execution pipeline

Developer

- Designed goal decomposition, importance-weighted memory decay, DAG execution visualisation, and Playwright verification into an end-to-end prompt-to-verified-task loop at 35% lower token cost.

AdityaOS - interactive 3D desktop portfolio

Solo build - live at adityabalaji.vercel.app

- Created an installable, mobile-ready desktop shell with Three.js/WebGL, research and project windows, local-first interactions, games, presence, and an adapted VS Code portfolio workspace.

LEADERSHIP, EXPERIENCE & SERVICE

MalkansView - market research intern

3 months, approximately 35 hrs/week

- Analysed company fundamentals against live market data and followed research through to the investment decisions the firm made.

Quantitative Finance Club - founder

25-30 members

- Designed a 13-week curriculum on markets, valuation, and quantitative methods; led Future Lab, a 150-company simulated market with intrinsic-value scoring, Ornstein-Uhlenbeck price dynamics, and timeline branching.

Sports Analysis Club - co-founder and co-host

Student-led community

- Built a consistent space for students to explore sports, data, and decision-making beyond formal coursework.

Rural Education Program & Fund Your Dream - volunteer

Teacher / fundraiser

- Taught foundational Mathematics and English to 50+ rural children, adapting delivery to student needs; raised INR 25,000+ for study kits, clothing, and footwear.

International Award for Young People (IAYP)

Bronze and Silver Awards

- Completed community service, football goalkeeping, and a 26-week quantitative research skills project focused on AI, machine learning, and quantitative trading.

TECHNICAL

Languages	Python, TypeScript/JavaScript, SQL, R
Research	Econometrics (IV, DiD, RD, event studies), GARCH/HAR volatility, factor construction, causal ML, Monte Carlo, agent-based simulation
Engineering	FastAPI, Next.js/React, PostgreSQL, pgvector, LangGraph, Prometheus, Docker, Numba, Playwright, Three.js/WebGL, Git, Vercel
Data	yfinance, FRED, Ken French, GBIF, NASA MODIS, NOAA GHCND, FAOSTAT